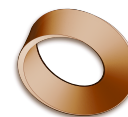


L2 – Energy Stores and Systems

P1 Energy • AQA Combined Science: Trilogy 8464



Physics 10 R and 10 P • Mr. Rawson

Date	Tuesday 1 September 2026	Room	PH1
10 R	P3, 11:15–12:10 (55 min)	Next	Thu 3 Sep, P2
10 P	P4, 12:15–13:05 (50 min)	Next	Wed 2 Sep, P2
Spec	P1.1.1 — aP1-1, aP1-2	Lesson	2 of 11
To print	None. Board work and books. 10 R = 11 students, 10 P = 19.		
Homework	Five everyday situations — name the store before, the store after, and the pathway between. Due next lesson.		

By the end of the lesson students can:

- name the eight energy stores and give an example of each;
- say what a *system* is, and describe how its stores change;
- distinguish a **store** from a **pathway**, and say why that matters in the exam.

Timing

Timed for the 50-minute period (10 P, P4). 10 R has five minutes more at P3 — spend them on the practice, not the exposition.

Time	Phase	What happens
0–5	Do Now	Board from L1: “the battery is used up”. Where did it go? Take two answers, leave it unresolved.
5–17	The eight stores	Kinetic, thermal, chemical, gravitational, elastic, electrostatic, magnetic, nuclear. One example each, from them not you.
17–29	Store vs pathway	The lesson’s hinge. Light, sound and electricity are <i>not</i> stores. Four pathways: mechanical work, electrical work, heating, radiation. Build the table on the board together.
29–44	The five AQA cases	In books, before-and-after stores for each: object thrown up; moving object hits an obstacle; object accelerated by a constant force; vehicle slowing; kettle boiling water. Circulate.
44–50	Consolidate	Exit question on the board. Set homework. Flag that from L3 we start <i>calculating</i> these changes.

Questions to ask

1. A ball is thrown straight up. What is the system, and which store grows as it rises?
2. Is “light energy” a store? What would you write instead?
3. A car brakes to a stop. The kinetic store has gone — gone where, exactly?
4. Kettle: name the store before, the store after, and the pathway between them.
5. Where does the energy come from that lets you throw the ball at all?

Common misconceptions

- **“Energy is used up.”** Carried over from L1 and it will not die quietly. Answer it explicitly today: dissipated to the thermal store of the surroundings, still there, no longer useful.
- **Store confused with pathway.** They will write “light energy”, “sound energy”, “electrical energy” as stores. AQA marks this wrong. This is the highest-value correction in the lesson.
- **“The system is everything.”** A system is what *you* choose to draw the box around. Make them state the boundary before describing changes.
- **Gravitational store belongs to the object alone.** It is the object *and* the Earth. Worth saying once, not labouring.

Differentiation

Support: give the eight stores as a card sort with picture prompts rather than asking for recall. Do the kettle case together before they work alone.

Extend: aP1-4 — sketch the five cases on a common scale, showing energy redistributed rather than lost. That is the groundwork for Sankey diagrams later.

After the lesson

- **10 R runs first, at P3.** Whatever falls flat, fix it before P4 — you get a second attempt with 10 P an hour later. Note the change here.
- Did the store/pathway distinction land? If not, L3 will produce “kinetic energy” answers with no store named. Reteach it in the Do Now.
- Mark both Class Log entries **Taught**, with the homework set.